DCIT 313: Introduction to Artificial Intelligence
University of Ghana

SleepWise

Sleep Quality Advisor Expert System

PROJECT BRIEF

Group	KENS
Group Lead	Ofori Kennedy Atta — 22110500
Knowledge Engineer	Ofori Kenneth Atta — 22173900
Course	DCIT 313: Introduction to Artificial Intelligence
Date	March 2026
Repository	DCIT313-GroupKENS-SleepWise-Expert-System

Academic Submission | DCIT 313 | University of Ghana

1. Project Overview

1.1 What SleepWise Is

SleepWise is a rule-based expert system that helps university students figure out what is hurting their sleep. It asks the student a set of questions, works out which causes are likely, and gives them a personalised report with specific steps to fix each problem.

The system runs on two parts working together. SWI-Prolog holds all the diagnostic logic, 29 cause rules, a four-level severity calculator, five clinical alert flags, and 29 recommendations. Python handles the interface, an adaptive Bayesian engine that decides which questions to ask next, and the session flow.

1.2 The Problem

The problem is that sleep issues rarely come from just one thing. A student might be drinking too much coffee, staying up past midnight, living in a noisy room, and feeling anxious about exams all at once. Most students can't tell which of those is actually the main problem. SleepWise works it out for them.

1.3 Why an Expert System

Sleep diagnosis is rule-based. Doctors and researchers have spent years building tools the PSQI, the ISI, the DASS-21, that turn answers to specific questions into diagnostic conclusions. That kind of reasoning fits Prolog well. We write a rule, and when the facts match, the conclusion fires. The intelligence stays in the knowledge base where a marker can read and verify it.

2. Objectives

2.1 Primary Objective

Build a working expert system that asks a student between 20 and 50 questions, identifies which of 29 causes are hurting their sleep, classifies the severity of the problem, and gives one specific recommendation per cause.

2.2 What the System Covers

Domain	Causes included
Lifestyle	Caffeine, diet, hydration, alcohol, drugs, medications
Psychological	Stress, anxiety, depression, burnout, rumination, perfectionism, self-control, resilience
Environmental	Noise, light, temperature, room comfort, neighbourhood safety
Behavioural	Screen use, sleep schedule, napping, sleep hygiene, sleep efficiency
Academic	Sleep duration, late-night studying, workload, exam pressure, performance impact

2.3 Success Criteria

- At least 25 Horn clause rules in sleepwise.pl covering all five domains.
- All diagnostic reasoning stays in the Prolog file — no if-else logic in Python.
- Same inputs always produce the same output.
- Every recommendation traces back to a specific fired Prolog rule.
- At least 10 structured positive and negative test cases pass without error.

3. Target Audience

SleepWise is built for full-time undergraduate students aged 18 to 25 at universities. Most live in shared accommodation, hostels, apartments, or family homes where noise, shared lighting, and limited control over the environment are common. Many rely on smartphones and laptops late into the night for both studying and entertainment.

These students know sleep matters but deprioritise it against deadlines and social commitments. They're comfortable with digital tools and more likely to use a technology-based system than visit a doctor. They need something free, fast, and direct and not generic advice, but a specific explanation of what is wrong and what to do about it tonight.

4. Scope of Work

4.1 System Files

File	Location	What it does
sleepwise.pl	knowledge_base/	All 29 Prolog cause rules, severity, complaints, clinical flags, recommendations
interface.py	interface/	Terminal interface, question display, Prolog bridge, session flow
bayes_engine.py	interface/	Bayesian engine — 29 causes, priors, likelihoods, severity scoring
question_bank.py	interface/	120 questions across 5 domains, 4 session variants (q20/q30/q40/q50)
session_manager.py	interface/	Session orchestration, adaptive question selection
trigger_rules.py	interface/	Cause trigger rules, cross-domain triggers, variant configuration

4.2 Session Variants

Students choose how many questions they want to answer. The four options are: Quick (20 questions), Standard (30), Detailed (40), and Full (50). The longer sessions reach more causes and include clinical questions about diagnosed disorders and panic attacks.

4.3 What the System Does Not Do

- No medical diagnosis or treatment plans.
- No wearable or real-time data — self-reported answers only.
- No graphical or web interface — terminal only.
- No persistent storage — each session starts fresh.

5. Timeline

Development ran from March 7 to March 12, 2026 — six active days. Documentation continued through to the submission deadline of March 21.

Day	Date	Work Completed	Who
1	Mar 7	GitHub repo created; folder structure set up; TA added as collaborator; notification email sent; README drafted; five domains and initial cause list agreed.	Both

Day	Date	Work Completed	Who
2	Mar 8	120-question bank designed with domain balance and evidence citations. question_bank.py completed.	Kennedy (bank) + Kenneth (cause design)
3	Mar 9	sleepwise.pl built: 13 Horn clause cause rules, severity logic, domain mapping, 13 recommendations, reset predicate. Tested in SWI-Prolog shell before integration.	Kenneth
4	Mar 10	bayes_engine.py built: 29 causes, evidence-based priors, likelihood tables, correlation correction, four-layer diagnosis. trigger_rules.py built.	Kennedy
5	Mar 11	session_manager.py and interface.py built. Full session loop working end-to-end with Prolog bridge.	Kennedy
6	Mar 12	Integration testing across all four variants. Critical bug fixes applied. Knowledge base expanded to 29 causes. Test scripts written — 267/267 pass.	Both
7–15	Mar 13–21	All four /docs documents completed and committed. README finalised. Submission prepared.	Both

6. Budget and Resources

SleepWise costs nothing to build. All tools are free and open source.

Tool / Resource	Purpose
SWI-Prolog	Logic engine — stores and processes all rules
Python 3.x + pyswip	Interface, Bayesian engine, Prolog bridge
VS Code	Code editor for both Prolog and Python files
Git & GitHub	Version control and submission platform
PSQI, ISI, DASS-21, CDC, AASM, Mbous 2022, Okano 2019, Wang 2023, Bousgheiri 2024, Stores 2023	Clinical thresholds, prevalence data, prior probabilities, rule justification

Total financial cost: GHS 0.00

7. Team and Roles

Name & ID	Role	Responsibilities
Ofori Kennedy Atta 22110500	Group Lead + Python Developer	question_bank.py, bayes_engine.py, trigger_rules.py, session_manager.py, interface.py; GitHub setup; integration testing; documentation lead.
Ofori Kenneth Atta 22173900	Knowledge Engineer + Prolog Developer	sleepwise.pl — all 29 Horn clause cause rules, severity logic, domain mapping, recommendation rules; knowledge engineering report; Prolog testing.

Shared Responsibilities

- Joint review of all Prolog rules for logical accuracy and domain coverage.
- Integration testing — verifying that Python and Prolog produce consistent results.
- All four /docs documents reviewed by both members before submission.
- Active individual GitHub commit histories maintained throughout.

8. Research Basis

Every rule in the knowledge base, every prior probability, and every likelihood weight traces back to a cited source. The key findings we built on:

- Over 60% of university students sleep fewer than six hours per night during term (NSF, CDC).
- Psychological stress is the strongest predictor of poor sleep — accounts for about 35% of variance in student populations (Wang 2023).
- 72% of African students use screens before bed. Screen use suppresses melatonin production (Bousgheiri 2024).
- Caffeine consumed after 2pm measurably delays sleep onset because its half-life is five to seven hours (NSF).
- Inconsistent sleep schedules — especially weekend catch-up sleeping — disrupt the circadian rhythm the same way jet lag does (AASM, Okano 2019).
- Burnout and perfectionism affect about 40% of university students and are strongly linked to rumination and delayed sleep onset (Stores 2023).
- In the Ghanaian context, depression prevalence in students ranges from 35% to 55%, with employment and financial stress as compounding factors (Mbous 2022).

Why Prolog Works Here

Sleep diagnosis is IF-THEN reasoning. Doctors do not run probability calculations — they ask questions and apply thresholds. If a student drinks high amounts of caffeine in the afternoon, that is a cause. Prolog is built for exactly that. We write the rules, and the inference engine applies them. No guesswork.

9. Constraints and Risks

9.1 Known Constraints

Constraint	Detail
Self-reported data only	The system cannot verify actual sleep data. Self-report bias corrections in the likelihood tables partially help, but underreporting of sensitive conditions is unavoidable.
Static knowledge base	Rules cannot update automatically. Adding new causes requires editing sleepwise.pl directly.
No persistent storage	Each session starts fresh. No cross-session tracking.
Terminal only	No graphical interface. The system runs in a command line window.

Constraint	Detail
Naive Bayes independence	Questions are not fully independent. Correlation pair corrections reduce the accuracy impact from about 3% to under 1%.

9.2 Risk Register

Risk	Likelihood	Impact	Mitigation
Prolog predicate format mismatch — Python asserts a fact in the wrong format, Prolog query returns empty	Medium	High	Assert facts as predicate(value) exactly matching sleepwise.pl structure. Verify with [Prolog ✓] tags in output during testing.
Bayesian engine fires causes for healthy students	Low	Medium	Anchor veto suppresses cause firing when 75%+ of anchor answers indicate healthy sleep.
Clinical flags triggering on insufficient evidence	Low	High	All five flags require specific multi-condition combinations. Each tested with dedicated negative profiles.
Turnitin similarity above 15%	Low	High	All documentation written in original language. Evidence paraphrased with full citations.

10. Deliverables

Deliverable	Location	Status
sleepwise.pl	knowledge_base/	Complete — 29 cause rules, severity, complaints, clinical flags, 29 recommendations
interface.py + supporting modules	interface/	Complete — full session pipeline working across all 4 variants
project_brief.pdf	docs/	Complete
knowledge_engineering_report.pdf	docs/	Complete — 10 sources, full rule-to-source mapping
test_cases.pdf	docs/	Complete — 267 test cases, all PASS
project_report.pdf	docs/	Complete
README.md	root	Complete

11. Closing Statement

SleepWise is a working expert system that combines SWI-Prolog Horn clause inference with a Naive Bayesian adaptive engine. It covers 29 sleep disruption causes, 120 questions, and four session variants. Every reasoning step is explainable. Every recommendation traces back to a specific Prolog rule. Every prior probability cites a source.

The system satisfies the KBS separation requirement of DCIT 313 by design — all intelligence is in sleepwise.pl, Python handles only the interface. The full test suite of 267 cases passes. All documentation is in /docs. The system is functional and ready for submission.

> **Definition of Done:** SleepWise is ready for submission when all four /docs documents are in place, both members have active commit histories, the TA is a confirmed collaborator, all test cases pass, and all documentation scores below 15% on Turnitin.

End of Project Brief — Group KENS | DCIT 313 | University of Ghana | March 2026